

$n=2$ :  $f(x) = y^2 - x^3$   $\mu=2$  ( $n=1$  case: base example in Iversen's table)

Gradient ideal:  $I_{\nabla f} = \langle x, y \rangle \Rightarrow$  Local algebra:  $\mathcal{O}_f = \mathbb{C}[[x, y]] / I_{\nabla f}$   
 $= \langle x^2, y \rangle$   
 $= \langle 1, x \rangle$   
 $\underline{t} = (0, b)$

Miniversal deformation:  $F_{\underline{t}}(x) = y^2 - x^3 - \underline{a}x - b$   
Crit. pts:  $x_{\pm} = \pm \sqrt{\frac{a}{3}}$   $\Rightarrow$  Discriminant  $\Delta = \{ (a, b) \in \mathbb{C}^2 : 4a^3 + 27b^2 = 0 \}$   
 $y_{\pm} = 0$

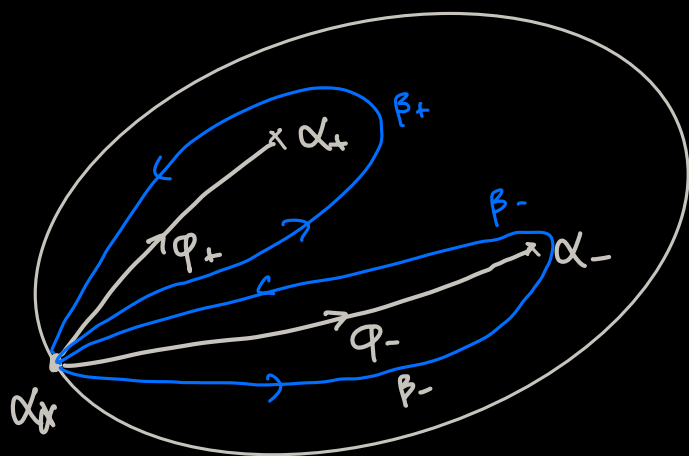
Picard-Lefschetz:

$f(x, y) = x^3 - y^2 \rightarrow \tilde{f}_{\underline{a}}(x, y) = y^2 - x^3 - \underline{a}x$  Morse fl., 2 Morse singularities  $\alpha_{\pm}$

$\tilde{f}_{\underline{a}}(\alpha_{\pm}) = \alpha_{\pm}$  critical values

So the situation is

$\tilde{f}: \mathbb{C}^2 \rightarrow \mathbb{C}$ , with



$\gamma_{\pm}$ : distinguished paths

$\tilde{f}^{-1}(\alpha_x)$  is  $\Sigma$ ,

as we follow along the path  $\gamma_{\pm}$  two branch pts collide and either  $\gamma_1$  or  $\gamma_2$  become trivial



$$Q = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

(cay. matrix of  $A_1$  given)

vanishing cycles

## Picard - Lefschetz monodromies:

$$h_i(\Delta_j) = \Delta_j - \underset{\substack{\uparrow \\ (-1)^{n(n+1)/2}}}{Q_{ji}} \Delta_i$$

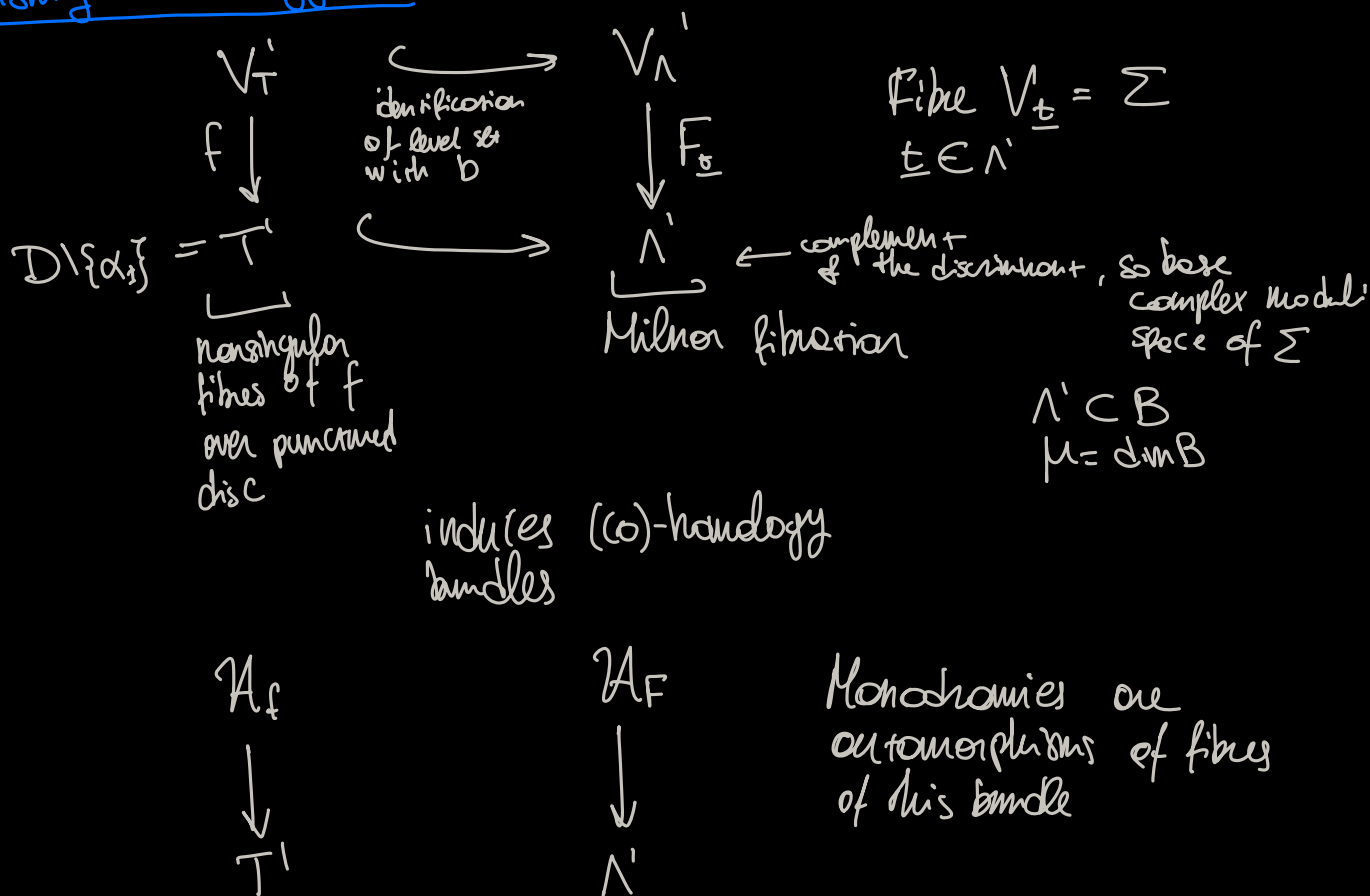
$$\begin{aligned} h_1(\Delta_1) &= \Delta_1 \\ (\text{no sum}) \quad h_1(\Delta_2) &= \Delta_2 + \Delta_1 \\ h_2(\Delta_1) &= \Delta_1 - \Delta_2 \end{aligned}$$

$$h_1 = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad h_2 = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}$$

monodromies along  $\beta_{\pm 1}$

Classical monodromy:  $h_{\alpha} = h_1 h_2 = \begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix}$

## Vanishing cohomology bundle:



- $\omega$  holomorphic  $(n-1) = 1$ -form in  $\mathbb{C}^n \times \mathbb{C}^M \subset \text{Tot}(V_{\Lambda} \rightarrow \Lambda')$   
 $\Rightarrow \omega_{\pm} = \omega|_{V_{\pm} = \Sigma}$  is closed because  $d\omega \in H^{2,0}(\Sigma) = 0$
- $\omega_{\pm} \rightsquigarrow [\omega]_{\pm} \in H^{n-1}(V_{\pm}; \mathbb{C}) \rightsquigarrow$  section  $[\omega]$  of  $\mathcal{H}_F \rightarrow \Lambda'$   
period map

- global section  $\sim \oint_{\gamma} \omega_{\pm}$  is a holomorphic multivalued function on the base

Example: For  $y^2 = x(x-1)(x-u)$

$$\omega = \frac{dx}{y}, \quad \text{period map } \Pi: (\lambda, u) \mapsto \left( \oint_{\Delta_1} \omega, \oint_{\Delta_2} \omega \right)$$

$$\oint_{\Delta_1} \omega = \int_0^1 \frac{dx}{\sqrt{x(x-1)(x-u)}}$$

$$= \frac{1}{\sqrt{u}} \int_0^1 \frac{dx}{\sqrt{x(x-1)(x-u/u)}} \propto {}_2F_1\left(\frac{1}{2}, \frac{1}{2}, 0; u\right)$$

$${}_2F_1(\alpha, \beta, \gamma; z) \propto \int_0^1 dx x^{\beta-1} (1-x)^{\gamma-\beta-1} (1-zx)^{-\alpha}$$

So the period map is a holomorphic multivalued fn:  
Gauss hypergeometric. The monodromy of the singularity is encoded in the monodromy of the hypergeometric fn

- NB: when we approach the discriminant,  $u \rightarrow 0, \infty$   
 $\lambda \rightarrow 0, \infty$   
 $\lambda \rightarrow u$   
the periods go to  $0, \infty$ .

- This is a trivialization of the cohomology bundle

- $\omega = \frac{dx}{y}$  is also what is called the residue form

$$\Omega_2 = dy \wedge dx = df \wedge \omega = (2y dy + (\dots) dx) \wedge \frac{dx}{y}$$

$$\Rightarrow \omega = \Omega_2 / df \quad \text{notation in the book}$$

Not  
Weyl

Gauss-Manin Connection & Picard-Fuchs equations: idea: analytic description of GM connection

Theorem: The covariant derivation  $\nabla_t$  of the GM connection in the vanishing cohomology bundle  $\mathcal{H}_f^* \rightarrow T'$  is

$$\nabla_t[\omega] = [d\omega/df] \quad (\text{singular 1st order system } \mathcal{H}_f^*)$$

the monodromy of this can be canonically identified with the classical monodromy of  $f$ .

•  $(\omega_1, \dots, \omega_\mu)$  trivialization of cohomology bundle  $\mathcal{H}_f^* \rightarrow T'$   
 $\Rightarrow [\omega_i/df]$  geometrical sections

• Away from the discriminant,  $\nabla_t[\omega_i/df]$  is a geometric section.

• Isolated singularity:  $f^k \nabla_t[\omega_i/df]$  is a geometric section everywhere if  $f^k \in \mathbb{C}^*$

$$\Rightarrow \boxed{\begin{aligned} \nabla_t[\omega_i/df] &= \sum_{\dot{j}} p_{i\dot{j}}(t) [\omega_{\dot{j}}/df] \\ p_{i\dot{j}}(t) &\text{ meromorphic} \end{aligned}}$$

$(\Delta_1, \dots, \Delta_n)$

• Let  $\Delta(t)$  be the covariantly constant section of the homology bundle  $\mathcal{H}_f \rightarrow T'$ , and define

$$I_{i\dot{j}} = \oint_{\Delta_{\dot{j}}} \omega_i/df.$$

- Action of GM connection  $\equiv$  Picard-Fuchs eqs (of the trivialization  $\omega$ )

$$\begin{cases} \partial_{t_i} I = P_i I \\ \text{Flatness: } \partial_{t_i} P_j - \partial_{t_j} P_i + [P_i, P_j] = 0 \end{cases}$$

- The monodromy at  $\infty$  of this system is the classical monodromy hnf of the singularity  $f$   
 $\text{loop} \rightsquigarrow$  only local analysis around a singularity, set at  $t=0$   
 + only one  $t$

- Let  $\frac{dI}{dt} = AI$  be a system of diff. eqs with  $A$  meromorphic at  $t=0$ . The fundamental matrix of solution is

$$\star \quad \Phi = Q(t) t^R \leftarrow \begin{matrix} \text{const. matrix} \\ \uparrow \\ \text{holom. in } T' \end{matrix}$$

$$\left[ \begin{aligned} \text{Ex: } \frac{dI}{dt} &= \frac{A_{-1}}{t^2} I, \quad A_{-1} = G \Lambda_{-1}^{\leftarrow \text{diagonal}} G^{-1} \\ \Rightarrow \Phi &= G \exp\left\{-\frac{\Lambda_{-1}}{t}\right\} \quad \text{with essential sing. at } t=0 \\ &\quad (\rightsquigarrow \text{Stokes phenomenon}) \end{aligned} \right]$$

### Regular singularity

If  $Q$  is meromorphic at  $t=0$ .

- By gauge transform:  $I \mapsto Q^{-1} I \Rightarrow \frac{dI}{dt} = \frac{R}{t} I$  (simple pole)

$$R = G_0 \Lambda_0 G_0^{-1} \Rightarrow \Phi = G_0 \underbrace{\Psi(t)}_{1+O(t)} t^{\Lambda_0} t^W \quad W \text{ nilpotent matrix} \\ \text{diag}(\alpha_1, \dots, \alpha_n) \quad W=0 : \alpha_i - \alpha_j \notin \mathbb{Z}$$

- This is the case relevant for PF equations of isolated singularities, because

Theorem (Nilsson):  $|\int \omega| \leq \text{const } |t|^{-N}$

Let  $0 < \text{arg } t < b$  Then  $\exists N$   
such that  $T'$ .

- This means there exist a basis of periods s.t.

$$d_t I = A(t) I, \quad \text{with } A(t) \text{ only simple poles at the singularities.}$$

- The classical monodromy is a matrix  $M = C_0^{-1} e^{2\pi i \Lambda_0} C_0$   
 $C_0 = \Phi(0)$

- The periods have (convergent) expansion

$$[\omega/df]|_{\delta} := \int_{\delta(t)} \omega/df_t = \left( \sum_{k=0}^{\infty} Q_{k, \alpha_i} t^k \right) t^{\alpha_i}$$

see Sect. 1.3 of

Fuchs 175 Kopelev

Nabokshenar  
Poincaré  
Transcendent  
RHP (pp. 175)

Matrix form  $I(t) = G_0 \left( \sum_{k=0}^{\infty} A_k t^k \right) t^{\Lambda_0} t^W$  Nilpotent matrix in resonant case

- In the case of singularity theory,  $\alpha_i \in \mathbb{Q}$

$$\Rightarrow \det(hx - \lambda E) = \prod_j P_j \quad \leftarrow \text{cyclotomic polynomial:} \\ P_j(\lambda) = \prod_{\substack{k=1 \\ \gcd(k,j)=1}}^j (\lambda - e^{2\pi i k/j})$$

$\Rightarrow h^N - 1_\mu$  is nilpotent for some  $N$

• The monodromies are highly nongeneric



$$\oint_{\Delta_1} \omega = \int_0^1 \frac{dx}{\sqrt{x(x-1)(x-u)}} = \omega_1$$

$$\oint_{\Delta_2} \omega = \int_1^u \frac{dx}{\sqrt{x(x-1)(x-u)}} = \omega_2 + \text{periods of } d=y dx$$

$$\begin{cases} \partial_u I = P_u I \\ \partial_\lambda I = P_\lambda I \end{cases}$$

$$\partial_u P_\lambda - \partial_\lambda P_u = [P_u, P_\lambda]$$

$$I' = \begin{pmatrix} \oint_{\Delta_1} \lambda & \oint_{\Delta_2} \lambda \\ \oint_{\Delta_1} \omega & \oint_{\Delta_2} \omega \end{pmatrix}$$

averaging over the reg

System of Picard-Fuchs equations

• The PF eqs are simply hypergeometric eq. with special  $\rho$ 's  
(recall periods were hypergeometric functions)

## BONUS: HOW TO COMPUTE PF EQUATIONS (in scalar form)

① Start from holomorphic  $(n-1)$ -form  $\omega = \frac{dx}{2y}$ , compute derivatives

$$\eta = \omega'' + A\omega' + B\omega \stackrel{!}{=} d\xi$$

$$\frac{d\omega}{d\sigma} = -\frac{x}{4y^3}$$

$$\frac{d^2\omega}{d\sigma^2} = \frac{3x^2}{8y^5}$$

$$x^3 + ax + b$$

↓

$$\frac{d^2\omega}{d\sigma^2} + A \frac{d\omega}{d\sigma} + B\omega = \frac{3x^2 - 2Ax y^2 + B y^4}{8y^5} dx$$

$$= \frac{P_6(x)}{y^5} dx$$

② Subtract linear combinations of  $d\left(\frac{x^k}{y^3}\right)$   
 $k=0, \dots, 4$   
until we have

$$\eta \mapsto \frac{P_1(x)}{y^5} dx, \text{ depending on } A, B.$$

Now we can set  $P_1(x) \equiv 0$  by fixing  $A, B$

$$\Rightarrow \boxed{\text{PF EQ: } \frac{d^2\omega}{d\sigma^2} + A \frac{d\omega}{d\sigma} + B\omega = 0}$$

Some approach holds more generally.